• **LAMP—LOW COOLANT ILLUMINATION WITH COOLANT AT NORMAL OPERATING LEVEL—VEHICLES WITH 2.5L ENGINE—SERVICE TIP** • **COOLING SYSTEM—LOW COOLANT LAMP ILLUMINATION WITH COOLANT AT NORMAL OPERATING LEVEL—VEHICLES WITH 2.5L ENGINE—SERVICE TIP**	**Article No. 00-2-4**

MERCURY: 1999-2000 COUGAR

ISSUE

The Low Coolant lamp may illuminate even though the coolant is at the normal operating level, on some vehicles. This may be due to either a marginal coolant level, a failed coolant level sensor, a sticking float, or a wiring issue.

ACTION

Refer to the following System Operation and Service Procedure to ensure the system is functioning as intended.

SYSTEM OPERATION

- The low coolant lamp will illuminate briefly to prove-out the lamp when the ignition key is first turned to On. It should normally turn off after 5 seconds.
- The lamp will illuminate any time a low coolant level signal is sent to the Hybrid Electronic Cluster (HEC) while driving. After the initial 5 second lamp prove-out, the HEC must see this low coolant level signal for greater than 33 seconds before the warning lamp will be illuminated. This is to prevent normal coolant level fluctuations which occur while driving from inadvertently activating the lamp.
- Anytime the lamp does turn on (after the 5 second prove-out) the ignition key must be cycled to the Off position before the lamp will be turned off again.
- The electrical contacts in the low coolant sensor are closed (warning lamp is off) when the coolant level is in the normal range and open (turns lamp on) when it drops below the MIN mark on the expansion tank.
- The sensor is not in direct contact with the coolant itself, but magnetically monitors the position of a float which is integrated into the expansion tank.

SERVICE PROCEDURE

If the lamp remains illuminated after the 5 second prove-out period:

NOTE
IT IS VERY IMPORTANT THAT THE COOLANT LEVEL BE CHECKED WITH A COLD ENGINE. CHECKING WHEN WARM OR HOT WILL RESULT IN AN INACCURATE COOLANT LEVEL!

NOTE
COOLANT LEVEL BELOW THE MAX LINE (WHEN CHECKED WITH A COLD ENGINE) IS THE MOST COMMON CAUSE FOR INTERMITTENT ILLUMINATION OF THE LAMP!

1. Check the coolant level in the expansion tank on the RH inner fender. Ensure the coolant level is at the MAX line, not just between MIN and MAX. Operating with the coolant level anywhere between the MIN and MAX line is acceptable to maintain proper engine operation. However, under certain operating conditions such as when parked on an incline, it is possible for the coolant level to temporarily drop below the MIN line in the vicinity of the coolant level sensor. This may result in the lamp illuminating when the engine is started. It is allowable to overfill the coolant level to a maximum of 10mm (0.375″) above the MAX line, and is advisable to do so if the vehicle is regularly parked on an incline.
2. If the coolant level is at or up to 10mm (0.375″) above the MAX line and the low coolant lamp remains illuminated for more than 5 seconds when the key is first turned on, determine if the low coolant level sensor or a sticking float in the expansion tank is the cause, or if there is a wiring concern:

a. Turn the ignition key off. Disconnect C812 at the low coolant sensor and connect a jumper wire across the two terminals in the harness connector.

b. Turn the key on and watch the lamp.
   - If the lamp now goes out after 5 seconds, go to Diagnostic/Repair Procedure A.
   - If the lamp continues to remain illuminated after 5 seconds, go to Diagnostic/Repair Procedure B.

DIAGNOSTIC/REPAIR PROCEDURE "A"

If the lamp now goes out after 5 seconds, the low coolant level sensor may have failed or the float in the expansion tank may be sticking. To determine which is at fault, remove the two screws which attach the expansion tank to the inner fender, unplug the harness connector from the sensor, then remove the sensor from the bottom of the expansion tank by twisting the sensor counterclockwise (it is NOT necessary to drain the coolant from the tank). Connect an ohmmeter to each of the two terminals on the sensor. The ohmmeter should read greater than 100 ohms (open circuit).

- If it reads less than 100 ohms, the sensor has shorted and should be replaced.
- If it reads greater than 100 ohms (open circuit), watch the ohmmeter while slowly sliding a magnet (such as on the end of a pocket screwdriver) along the length of the sensor. If at any time the ohmmeter now reads less than 100 ohms, this indicates that the sensor is responding properly to a magnetic signal (similar to that provided by the float in the expansion tank) and should not be replaced.

A reading less than 100 ohms indicates that the float inside of the expansion tank is operating improperly (sticking). The float is not serviceable, so the expansion tank should be replaced. If it does not read less than 100 ohms at any time as the magnet is moved along the sensor, then the sensor is not responding to the magnetic input and the sensor alone should be replaced.

DIAGNOSTIC/REPAIR PROCEDURE "B"

If the lamp stays on, use a voltmeter (NOT a test lamp!) to check for 12 volts from Circuit 8-WB9 (WH/GN) at the harness connector to the negative terminal of the battery.

- If 12 volts are present, then the ground circuit (black) to the second terminal is open.
- If 12 volts are not present, then there is an open in Circuit 8-WB9 (WH/GN) between the harness connector and the HEC.

PART NUMBER	PART NAME
F5RZ-17B649-A F8RZ-8A080-FD	Coolant Level Switch Overflow Tank

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 206000, 402000